

ITCCA Project 1

Faculty:	Information Technology
Module Code:	ITCCA4-34
Module Name:	Cloud Computing Architecture
Content Writer:	Dr. B. Maodza
Internal Moderation:	Community of Practice
Copy Editor:	Kyle Keens
Total Marks:	100
Submission Week:	Week 6 B2

Name and Surname:	Keatlegile Ntsie
Student No.:	EDUV9197834
Date:	06 August 2025

Project Deliverable 1: Project Proposal E-Commerce Platform.

Project Title

Ubuntu Crafts Cooperative E-commerce platform

Description

A catalogue e-commerce platform design to build a resilient, cost-effective and a user-friendly experience over both desktop via web application and a mobile application for user devices for Ubuntu Crafts. Allowing potentials and current clients access to products such as pottery, woven baskets, traditional clothing, and organic skincare items made by local artisans.

Problem Statement

Ubuntu Crafts currently sells its products through local crafts markets, informal traders and small retail outlets. This as a result limits the company's availability to locals and hinders them from reaching communities outside of the area they based in Limpopo.

Objectives

The aim is to build a secure and scalable e-commerce platform that will help the organisation grow its market reach and support economic development. The platform will be developed using AWS cloud services and will focus on:

- Creating a secure online store with user login, encrypted transactions, and strong data protection.
- Designing the system to handle more traffic and transactions without affecting performance.
- Automating order processing to cut down on manual work and reduce mistakes.
- Making the site accessible on desktops, tablets, and mobile devices.
- Integrating payment systems for a simple and quick checkout process.
- Using AWS Free Tier to keep costs low during testing and deployment.

The goal is to provide a reliable, easy-to-use platform that can grow with the needs of the business.

Scope

List of the AWS services and functionalities that will be used:

- **Amazon S3** – To host static website assets such as images, product pages, and stylesheets.
- **Amazon CloudFront** – A Content Delivery Network (CDN) for fast global delivery of content.
- **AWS Amplify** – For frontend hosting and easy integration with backend services.
- **Amazon RDS (Free Tier: MySQL/PostgreSQL)** – To store product details, orders, and customer information in a secure relational database.
- **AWS Lambda** – For serverless backend functions handling order processing, payments, and notifications.
- **Amazon API Gateway** – To manage secure communication between the frontend and backend.
- **Amazon Cognito** – For user authentication and account management (sign-up, login, and password recovery).
- **Amazon SES/SNS** – For sending order confirmations, marketing emails, and notifications.
- **Amazon CloudWatch** – To monitor and log platform performance.
- **AWS IAM** – For access control and secure resource management.

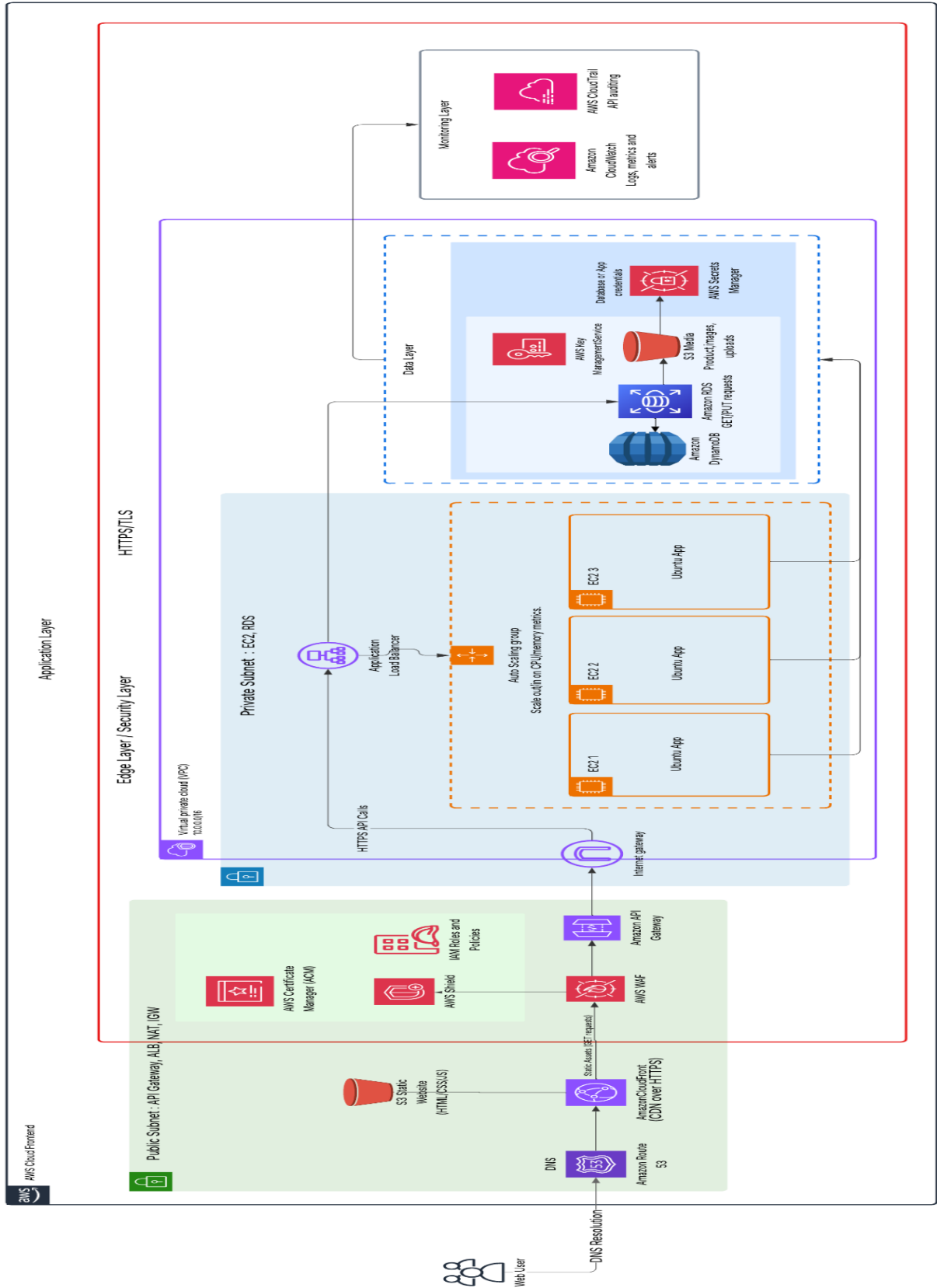
Timeline

A project timeline with key milestones and expected deadlines.

Phase	Activities	Duration	Deadline
Week 1 – Project Setup	AWS account setup, IAM roles, requirements gathering, initial architecture design	2 days	11/08/2025
Week 2 – Backend Setup	Configure RDS database, set up Lambda functions, API Gateway endpoints, Cognito user pool	5 days	16/08/2025
Week 3 – Frontend Dev	Develop UI with AWS Amplify hosting, integrate API calls, setup S3 for static content	6 days	22/08/2025

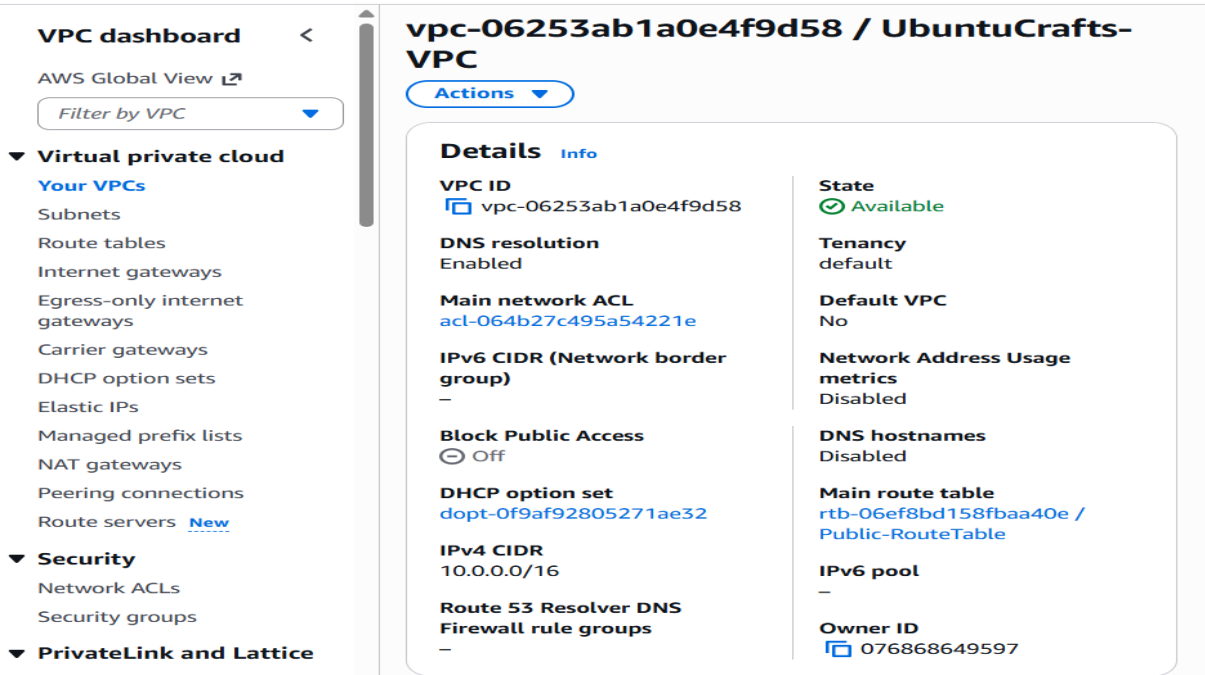
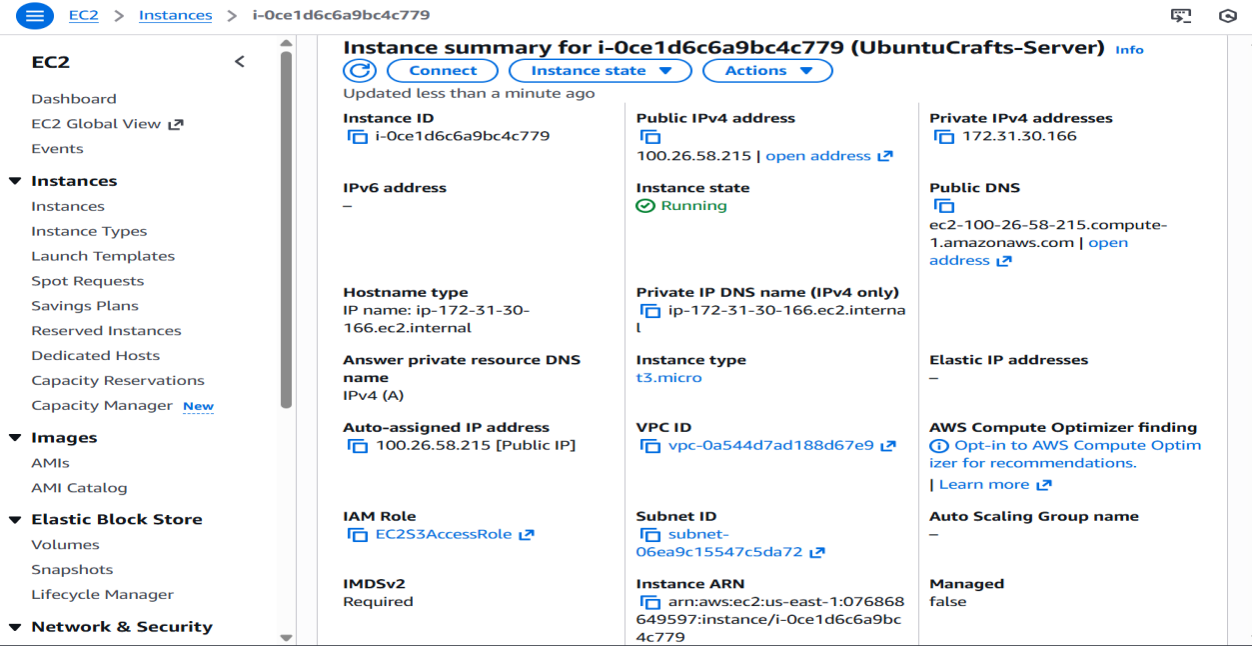
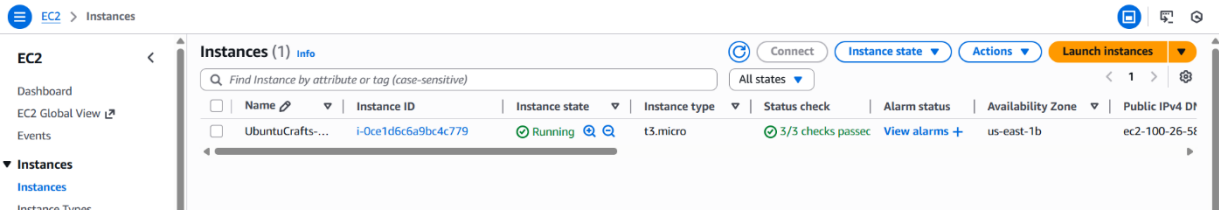
Week 4 – Integration	Integrate payment gateway, set up SES/SNS for notifications, connect frontend to backend services	4 days	26/08/2025
Week 5 – Testing	Functional testing, security checks, performance monitoring, bug fixing	5 days	01/09/2025
Week 6 – Deployment	Final deployment, documentation with screenshots, terminate unnecessary services	2 days	03/09/2025

Deliverable 2: Architecture Design



Project Deliverable 3: Implementation

Section A



VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

subnet-0120a0fd722b21cfe / Private-Subnet

Actions

Details

Subnet ID

subnet-0120a0fd722b21cfe

IPv4 CIDR

10.0.2.0/24

Availability Zone

use1-az4 (us-east-1b)

Network ACL

acl-064b27c495a54221e

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

Subnet ARN

arn:aws:ec2:us-east-1:076868649597:subnet/subnet-0120a0fd722b21cfe

Available IPv4 addresses

251

Network border group

us-east-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

State

Available

IPv6 CIDR

-

VPC

vpc-06253ab1a0e4f9d58 | UbuntuCrafts-VPC

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

076868649597

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-06ef8bd158fbaa40e | Public-RouteTable

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

VPC > Subnets > subnet-0af976f5f751ed668

Actions

Details

Subnet ID

subnet-0af976f5f751ed668

IPv4 CIDR

10.0.1.0/24

Availability Zone

use1-az4 (us-east-1b)

Network ACL

acl-064b27c495a54221e

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

Subnet ARN

arn:aws:ec2:us-east-1:076868649597:subnet/subnet-0af976f5f751ed668

Available IPv4 addresses

251

Network border group

us-east-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

State

Available

IPv6 CIDR

-

VPC

vpc-06253ab1a0e4f9d58 | UbuntuCrafts-VPC

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

076868649597

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-06ef8bd158fbaa40e | Public-RouteTable

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

VPC > Internet gateways > igw-0ccf4f5a3dd09ba10

Actions

Details

Internet gateway ID

igw-0ccf4f5a3dd09ba10

VPC ID

vpc-06253ab1a0e4f9d58 | UbuntuCrafts-VPC

State

Attached

Owner

076868649597

Tags (1)

Manage tags

Search tags

< 1 > ⚙

Key

Value

Name

UbuntuCrafts-IGW

☰

VPC > Route tables > rtb-06ef8bd158fbaa40e

ⓘ

🔍

🔄

AWS Global View 🌐

Filter by VPC ▾

▼ Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers [New](#)

▼ Security

Network ACLs

Security groups

▼ PrivateLink and Lattice

Getting started

rtb-06ef8bd158fbaa40e / Public-RouteTable

Actions ▾

Details [Info](#)

Route table ID

🔗 rtb-06ef8bd158fb
aa40e

Main

🔗 Yes

Explicit subnet associations

subnet-
0af976f5f751ed668 /
Public-Subnet

Edge associations

-

VPC

vpc-
06253ab1a0e4f9d58 |
UbuntuCrafts-VPC

Owner ID

🔗 076868649597

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Both ▾

Edit routes

🔍 Filter routes

< 1 > ⚙️

Destination ▾	Target ▾	Status ▾	Propag... ▾	Route Origin ▾
0.0.0.0/0	igw-0ccf4f...	🟢 Active	No	Create Route
10.0.0.0/16	local	🟢 Active	No	Create Route Table

☰

VPC > Security Groups > sg-0659ab17517c3e673 - UbuntuCrafts-EC2-SG

🔍

🔄

NAT gateways

Peering connections

Route servers [New](#)

▼ Security

Network ACLs

Security groups

▼ PrivateLink and Lattice

Getting started

Endpoints

Endpoint services

Service networks

Lattice services

Resource configurations

Resource gateways

Target groups

Domain verifications [New](#)

▼ DNS firewall

Rule groups

sg-0659ab17517c3e673 - UbuntuCrafts-EC2-SG

Actions ▾

Details

Security group name

🔗 UbuntuCrafts-EC2-
SG

Security group ID

🔗 sg-0659ab17517c
3e673

Description

🔗 Security group for
EC2 backend

VPC ID

vpc-
06253ab1a0e4f9d58

Owner

🔗 076868649597

Inbound rules count

2 Permission entries

Outbound rules count

0 Permission entries

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (2)

🔄

Manage tags

Edit inbound rules

🔍 Search

< 1 > ⚙️

☐	Name ▾	Security group rule ID ▾	IP version ▾	Type
☐	-	sgr-02c35b187a4dd7543	IPv4	HTTP
☐	-	sgr-09fead4aa8c481b6e	IPv4	SSH

DB Identifier	Status	Role	Engine	Region	Size	Recommendations	CPU	Current activity	Maintenance	VPC	Multi-AZ
database-1	Available	Instance	PostgreSQL	us-east-1f	db.t4g.mi...		3.92%	0.00 sessions	none	vpc-0a54...	No
ubuntucraftsdb	Creating	Instance	PostgreSQL	us-east-1b	db.t4g.mi...		-		none	vpc-0a54...	No

ubuntucraftsdb

Refresh Modify Actions

Summary

DB identifier
ubuntucraftsdb

CPU
3.81%

Recommendations

Status
Available

Class
db.t4g.micro

Role
Instance

Current activity
0.00 sessions

Engine
PostgreSQL

Region & AZ
us-east-1b

Connectivity & security Monitoring Logs & events Configuration Zero-ETL integrations More

Connectivity & security

Endpoint & port

Endpoint
ubuntucraftsdb.ccfu86q063er.us-east-1.rds.amazonaws.com

Port
5432

Networking

Availability Zone
us-east-1b

VPC
vpc-0a544d7ad188d67e9

Subnet group
default-vpc-0a544d7ad188d67e9

Subnets
subnet-09f85c1027e556c88
subnet-0d06f154c515c1e02
subnet-0a9896c23fa6761d8
subnet-0b1dac2c990856c77
subnet-024e493b65bc720b2
subnet-06ea9c15547c5da72

Network type
IPv4

Security

VPC security groups
rds-sg (sg-0613e58bef9ff7952)
Active

Publicly accessible
Yes

Certificate authority
rds-ca-rsa2048-g1

Certificate authority date
May 26, 2061, 01:34 (UTC+02:00)

DB instance certificate expiration date
November 11, 2026, 18:33 (UTC+02:00)

General purpose buckets (1) Info

Buckets are containers for data stored in S3.

Find buckets by name

Refresh Copy ARN Empty Delete Create bucket

1

Name	AWS Region	Creation date
ubuntu-crafts-storage	US East (N. Virginia) us-east-1	October 19, 2025, 08:00:36 (UTC+02:00)

☰

KMS > Customer managed keys

ⓘ | 🔒

✔ Success

Your AWS KMS key was created with alias **ubuntu-crafts-key** and key ID **da642277-f5b8-4034-bbce-37fbff795455**.

View key ✕

Customer managed keys (1)

Key actions ▾ Create key

🔍 Filter keys by properties or tags

< 1 > ⚙️

☐	Aliases ▾	Key ID ▾	Status	Key type ▾	Key spec ⓘ	Key usage
☐	ubuntu-crafts-key	da642277-f5b8...	Enabled	Symmetric	SYMMETRIC_DE...	Encrypt and dec...

☰

CloudWatch > Metrics

ⓘ | 📁 | 🔒

Metrics ⓘ

3h 1d 1w 📅 UTC timezone ▾ Actions ▾ Investigate ▾

CPUUtilization 🔗

Line ▾ ↺ ▾

Add math ▾ Add query ▾

Browse Multi source query **Graphed metrics (1)** Options Source

Add dynamic label ▾ ⓘ

Average ▾ 5 minutes ▾ Clear graph

☒	Label	Details	Statistic	Period
☒	CPUUtilization 🔗	EC2 • CPUUtilization • InstanceId: i-0ce1d 🔗	Average ▾	5 minu... ▾

☰

CloudWatch > Alarms

ⓘ

CloudWatch <

Favorites and recents ▶

Dashboards

▶ AI Operations [New](#)

▼ Alarms ⚠️ 0 ✔️ 0 ⌛ 0

In alarm

[All alarms](#) [New](#)

Billing

▶ Logs

▼ Metrics

[All metrics](#) [New](#)

✔ Successfully created alarm EC2-HighCPU-Alarm.

View alarm ✕

Alarms (1)

☐ Hide Auto Scaling alarms Clear selection ↺ Create composite alarm Actions ▾

Create alarm

🔍 Search

Alarm state: Any ▾ Alarm type: Any ▾ Actions status: Any ▾

< 1 > ⚙️

☐	Name ▾	State ▾	Last state update (UTC) ▾	Conditions
☐	EC2-HighCPU-Alarm	⌛ Insufficient data	2025-11-12 02:23:36	CPUUtilization >= 1000 5 minutes

